

Five Practical Transformer Examples – Different Types

Example 1: No-load Test

A single-phase transformer rated at 2 kVA, 230/115 V is tested on the 230 V side. During the no-load test, the wattmeter reads 60 W, and the ammeter reads 0.5 A. Find the core loss components and no-load power factor.

Given: $V_1 = 230 \text{ V}$, $I_0 = 0.5 \text{ A}$, $P_{\text{core}} = 60 \text{ W}$

$$I_w = P_{\text{core}} / V_1 = 60 / 230 = 0.2609 \text{ A}$$

$$I_m = \sqrt{(I_0^2 - I_w^2)} = \sqrt{(0.5^2 - 0.2609^2)} = 0.426 \text{ A}$$

$$\text{No-load power factor} = \cos\phi_0 = I_w / I_0 = 0.2609 / 0.5 = 0.522$$

Example 2: Short-circuit Test

A 5 kVA, 230/115 V transformer is tested by short-circuiting the secondary. The wattmeter reads 150 W at 10 V and 21.74 A on the primary. Find equivalent resistance, reactance, and impedance referred to the primary.

$$R_{\text{eq}} = P_{\text{sc}} / I_1^2 = 150 / (21.74^2) = 0.317 \Omega$$

$$Z_{\text{eq}} = V_{\text{sc}} / I_1 = 10 / 21.74 = 0.46 \Omega$$

$$X_{\text{eq}} = \sqrt{(Z_{\text{eq}}^2 - R_{\text{eq}}^2)} = \sqrt{(0.46^2 - 0.317^2)} = 0.33 \Omega$$

Example 3: Efficiency at Different Loads

A 10 kVA, 2300/230 V transformer has full-load efficiency of 94% at 0.8 PF lagging. Core loss = 150 W. Find efficiency at half load and 0.8 PF lagging.

$$\text{At full load: } P_{\text{out}} = 10,000 \times 0.8 = 8,000 \text{ W}$$

$$\text{Total loss} = (8,000 / 0.94) - 8,000 = 511 \text{ W}$$

$$\text{Copper loss} = 511 - 150 = 361 \text{ W}$$

$$\text{At half load: } \text{Cu loss} = (0.5)^2 \times 361 = 90.25 \text{ W}$$

$$\text{Efficiency} = \text{Output} / (\text{Output} + \text{Losses}) = 4,000 / (4,000 + 240.25) = 94.3\%$$

Example 4: Voltage Regulation at Different Power Factors

A 15 kVA, 2300/230 V transformer has $R_{\text{eq}} = 0.3 \Omega$ and $X_{\text{eq}} = 0.4 \Omega$ (referred to secondary). Find regulation at 0.8 PF lagging and 0.8 PF leading.

$$\text{At 0.8 lagging: } \Delta V = I (R_{\text{eq}} \cos\phi + X_{\text{eq}} \sin\phi) = 65.2 (0.3 \times 0.8 + 0.4 \times 0.6) = 26.08 \text{ V} \rightarrow 11.3\%$$

$$\text{At 0.8 leading: } \Delta V = I (R_{\text{eq}} \cos\phi - X_{\text{eq}} \sin\phi) = 65.2 (0.3 \times 0.8 - 0.4 \times 0.6) = 3.9 \text{ V} \rightarrow 1.7\%$$

Example 5: Parallel Operation of Transformers

Two transformers operate in parallel: T1 (kVA=50, impedance=5%) and T2 (kVA=25, impedance=4%). Find load sharing when supplying 60 kVA at unity PF.

Load shared is inversely proportional to impedance per kVA:

$$T1 \text{ load} = 60 \times (1/5) / (1/5 + 1/4) = 26.7 \text{ kVA}$$

$$T2 \text{ load} = 60 - 26.7 = 33.3 \text{ kVA}$$

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